PT.SUMITOMO ELECTRIC WINTEC INDONESIA

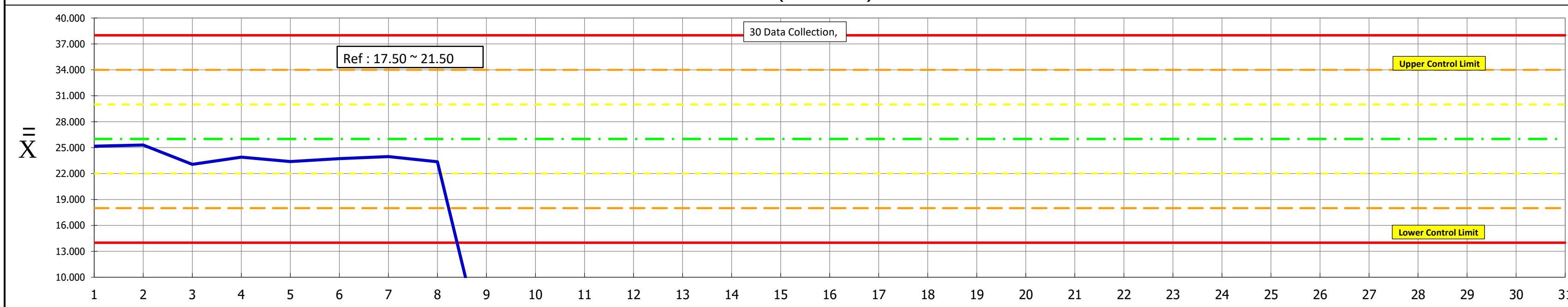
$\bar{X}$ - R CONTROL CHART

No. Dokumen	4-QAS-PP07-001
Revisi	00
Tgl Berlaku	2-Dec-16

MACHINE	SECTION	OPERATION	DATE CONTROL LIMITS CALC.	SPECIFICATION	COURSE NO.	CONTROL ITEM (Δ) YES ○ NO
DF Tester (Old)	QC	DF 250^0 C		min 12 max 40		
	MONTH	CHARACTERISTIC		SAMPLE SIZE / FREQUENCY	TYPE / SIZE : **D2DSW2 / 0.50**	
	May-25	Temperatur		n = 3	PIC : **DWI AS**	

$\bar{\bar{X}}$ = AVERAGE $\bar{X}$ = 26.000 U.C.L. = $\bar{\bar{X}} + A_2\bar{R}$ = 38.0000 L.C.L. = $\bar{\bar{X}} - A_2\bar{R}$ = 14.0000

AVERAGE (X BAR CHART)



SPECIAL CAUSES

- ☑ TEST 1: ANY POINT OUT OF CONTROL LIMIT
- ☑ TEST 2: 7 POINTS IN A ROW ON SAME SIZE OF CENTER LINE
- ☑ TEST 3: 6 POINTS IN A ROW ALL INCREASING OR DECREASING
- ☑ TEST 4: 14 POINTS IN A ROW, ALTERNATING UP & DOWN
- ☑ TEST 5: 2 OUT OF 3 POINTS > 2 STDEV FROM CENTER LINE (SAME SIDE)
- ☐ TEST 6: 4 OUT OF 5 POINTS > 1 STDEV FROM CENTER LINE (SAME SIDE)
- ☑ TEST 7: 15 POINTS IN A ROW WITHIN 1 STDEV OF CENTER LINE (EITHER SIDE)
- ☑ TEST 8: 8 POINTS IN A ROW > 1 STDEV FROM CENTER LINE (EITHER SIDE)

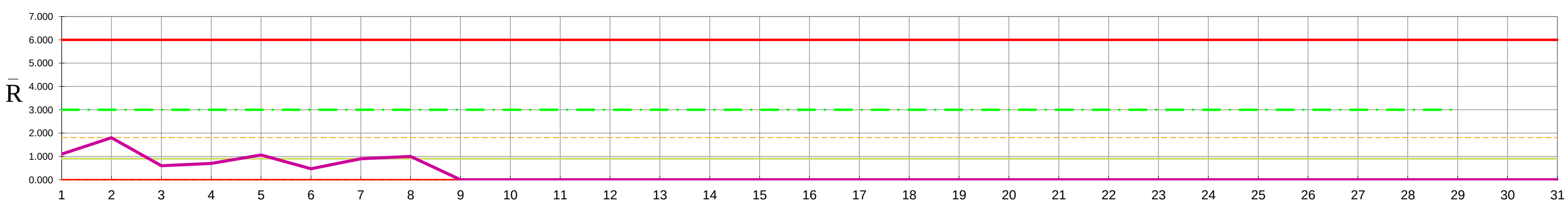
ACTION INSTRUCTION

1.
2.
3.
4.
5.

$\bar{R}$ = AVERAGE R = 3.000 U.C.L. = $D_4\bar{R}$ = 6.000 L.C.L. = $D_3\bar{R}$ = 0.00

Special Cause Marker Legend:
Test 1 ● Test 2 ◆ Test 3 ▲ Test 4 ■

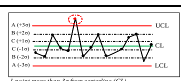
RANGES (R - CHART)



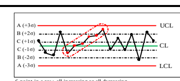
SUBGROUP SIZE	A_2	d_2	D_3	D_4
2	1.881	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.281
5	0.577	2.326	0	2.115
6	0.483	2.534	0	2.004
7	0.4193	2.704	0.0756	1.9244
8	0.3726	2.847	0.1361	1.8639

Special Cause

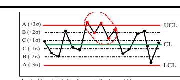
Titik berada di luar garis merah atau lebih dari 3 Ʊ



6 titik berada dalam satu urutan secara terus menerus, baik posisi naik ataupun turun



4 atau 5 titik lebih dari 1Ʊ dengan posisi yang sama dan titik keluar dari garis merah atau 3Ʊ



Date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
DATA 1	25.600	26.200	23.400	24.000	23.220	24.020	24.500	22.800																							
DATA 2	24.500	25.300	22.800	23.500	22.950	23.550	23.800	23.500																							
DATA 3	25.400	24.400	23.000	24.200	24.010	23.620	23.600	23.800																							
N	3	3	3	3	3	3	3	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AVERAGE, X	25.167	25.300	23.067	23.900	23.393	23.730	23.967	23.367	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
Range	1.100	1.800	0.600	0.700	1.060	0.470	0.900	1.000																							
Moving Range		0.002	0.768	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
SHIFT 1		DWI AS	DWI AS	DWI AS			DWI AS	DWI AS																							
SHIFT 2																															

CALCULATION

$\bar{\bar{X}}$ = 23.9863

$\bar{R}$ = 0.9538

s_{ST} = 0.5633 s_{LT} = 0.904

THE PROCESS MUST BE IN CONTROL BEFORE CAPABILITY CAN BE DETERMINED

$$C_{PK} = \text{MIN}\left[\frac{\bar{X} - LSL}{3s_{ST}} \quad \frac{USL - \bar{X}}{3s_{ST}}\right] = \text{MIN}\left[5.91 \quad 8.29\right] = 5.91$$

$$Cp = \left[\frac{USL - LSL}{6s_{ST}}\right] = \left[7.10\right]$$

CHECKED BY	APPROVED BY